Machine Learning-Based Cyber Attack Detection Using Security Event Data

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***Abstract*—The rapid increase in cyber incidents has made traditional rule-based security systems less effective in detecting modern and evolving threats. Organizations now generate large volumes of security event data, making manual monitoring inefficient and error-prone. Machine Learning offers the ability to automatically analyze security logs, identify suspicious behaviour, and support faster and more consistent threat detection. This study focuses on applying data-driven techniques to improve cyber attack identification using real-world cybersecurity event records. The objective of this research is to develop a Machine Learning-based approach for detecting and analysing cyber attacks using a structured cybersecurity dataset. The study aims to classify different attack types, examine attack severity patterns, and improve automated threat detection accuracy while reducing reliance on manual monitoring and human intervention. An experimental approach is followed using a cybersecurity event dataset containing attack logs, severity levels, network attributes, and response actions. The dataset is preprocessed through data cleaning, feature selection, and encoding of categorical variables. Machine Learning classification techniques are applied to identify patterns associated with various cyber attacks. Model performance is evaluated using standard metrics such as accuracy, precision, recall, and F1-score. The analysis demonstrates that Machine Learning models can effectively detect and classify cyber attacks, improve detection consistency, and reduce response time compared to manual methods. This study concludes that data-driven approaches enhance cybersecurity efficiency and provide scalable solutions for modern threat detection.**

Keywords—Machine Learning, Cyber Attack Detection, Cybersecurity Dataset, Threat Classification, Security Analytics

I. INTRODUCTION

The increasing reliance on digital systems and internet-based services has led to a rapid rise in cyber threats across various sectors, including finance, healthcare, education, and government infrastructure [1]. As organizations continue to generate and store large volumes of sensitive data, the risk of cyber attacks such as malware, phishing, distributed denial-of-service (DDoS), and ransomware has grown significantly. Traditional cybersecurity systems primarily rely on rule-based mechanisms and signature-based detection methods, which are effective in identifying known threats but often fail to detect new, evolving, or previously unseen attack patterns. This limitation creates a critical need for more intelligent and adaptive security solutions.

In recent years, Machine Learning has emerged as a powerful approach for enhancing cybersecurity systems. By leveraging data-driven techniques, Machine Learning models can analyze large-scale security data, identify complex patterns, and detect anomalies that may indicate malicious activity [2]. Unlike conventional systems, these models are capable of learning from historical data and improving their performance over time, making them well-suited for dynamic and unpredictable threat environments.

The ability to automate threat detection also reduces the dependency on manual monitoring and enables faster response to potential security incidents.

Cybersecurity event logs and structured datasets provide valuable insights into network activity, user behaviour, and system responses. These datasets often contain information such as attack types, severity levels, network attributes, and response actions, which can be utilized to train Machine Learning models for classification and prediction tasks [3]. Analyzing such data enables the identification of attack patterns and supports the development of intelligent systems capable of distinguishing between normal and malicious behaviour. However, challenges such as data preprocessing, feature selection, and handling imbalanced datasets must be addressed to ensure accurate and reliable model performance.

Despite the growing adoption of Machine Learning in cybersecurity, there remains a need for practical, implementation-based studies that demonstrate how these techniques can be applied effectively to real-world datasets. Many existing approaches focus on theoretical concepts without providing sufficient experimental validation. This research addresses this gap by applying Machine Learning techniques to a cybersecurity event dataset to detect and classify different types of cyber attacks. The study emphasizes data preprocessing, model development, and performance evaluation using standard metrics.

The main contribution of this work lies in developing a data-driven framework for cyber attack detection that leverages Machine Learning algorithms to improve detection accuracy and efficiency. The proposed approach aims to enhance automated threat analysis, reduce response time, and support scalable cybersecurity solutions. The remainder of this paper is organized as follows: Section II presents the problem statement and research objectives, Section III discusses the related work, Section IV describes the proposed methodology, Section V presents the implementation and results, and Section VI concludes the study with future directions.

II. PROBLEM STATEMENT

The increasing volume and complexity of cyber threats present significant challenges for traditional cybersecurity systems. Conventional security mechanisms rely heavily on rule-based detection and predefined signatures, which are effective only for known attack patterns. These systems often fail to detect new, evolving, or sophisticated cyber attacks such as zero-day exploits, advanced persistent threats, and polymorphic malware. As a result, organizations remain vulnerable to undetected intrusions and delayed responses.

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Additionally, the rapid growth of security event data generated from networks, systems, and applications makes manual monitoring and analysis inefficient and impractical. Security analysts face difficulties in identifying meaningful patterns within large datasets, leading to increased chances of human error and delayed threat detection. This creates a need for automated and intelligent systems capable of analyzing large-scale cybersecurity data efficiently [2].

Another major challenge lies in accurately classifying different types of cyber attacks while minimizing false positives and false negatives. High false alarm rates can overwhelm security teams, while missed detections can result in severe security breaches. Therefore, there is a need for a reliable and scalable approach that can improve detection accuracy and support timely decision-making.

This research addresses these challenges by proposing a data-driven Machine Learning-based approach for detecting and classifying cyber attacks using structured cybersecurity event data.

III. OBJECTIVES

The primary objective of this research is to develop an efficient Machine Learning-based system for detecting and classifying cyber attacks using a structured cybersecurity dataset. The study focuses on leveraging data-driven techniques to enhance automated threat detection and improve overall cybersecurity performance.
The specific objectives of this research are as follows:

- To analyze and preprocess cybersecurity event data for effective use in Machine Learning models
- To identify key features that contribute to accurate detection of cyber attacks
- To apply Machine Learning algorithms for classification of different types of cyber threats
- To evaluate model performance using metrics such as accuracy, precision, recall, and F1-score
- To improve detection efficiency while reducing false positives and false negatives

IV. RELATED WORK

The increasing complexity of cyber threats has led researchers to explore advanced techniques for improving threat detection and prevention. Traditional cybersecurity approaches, primarily based on signature matching and rule-based systems, have been widely used for identifying known threats. However, these methods are limited in their ability to detect unknown or evolving attack patterns, which has encouraged the adoption of Machine Learning techniques in cybersecurity applications [1],[2].

Several studies have demonstrated the effectiveness of Machine Learning algorithms in detecting cyber attacks by analyzing network traffic and system behavior. Supervised learning techniques such as Decision Trees, Support Vector Machines (SVM), and Random Forest have been widely used for classification tasks in intrusion detection systems [4]. These models are capable of learning patterns from labeled datasets and classifying network activity as normal or malicious with reasonable accuracy. Among these, Random Forest has shown strong performance due to its ability to handle high-dimensional data and reduce overfitting.

Deep Learning techniques have also gained attention in recent years for their ability to model complex patterns in large datasets. Neural Networks, including Convolutional Neural Networks (CNN) and Recurrent Neural Networks (RNN), have been applied to cybersecurity problems such as intrusion detection and malware classification [5]. These approaches are particularly useful for analyzing sequential data and identifying hidden relationships within network traffic. However, they often require large datasets and higher computational resources, which may limit their practical implementation in some environments.

Another important area of research focuses on anomaly detection, where models are trained to identify deviations from normal system behavior. Unsupervised learning methods such as clustering and autoencoders are commonly used for this purpose. These techniques are useful for detecting previously unseen attacks, including zero-day vulnerabilities, as they do not rely on labeled data. However, anomaly detection methods may produce higher false positive rates, which can affect their reliability in real-world applications.

Recent studies have also emphasized the importance of data preprocessing and feature selection in improving the performance of Machine Learning models. Techniques such as normalization, encoding of categorical variables, and removal of irrelevant features play a crucial role in enhancing model accuracy and reducing computational complexity [6]. Proper data preparation ensures that models can effectively learn from the dataset and produce reliable predictions.

Despite significant advancements, there are still challenges in implementing Machine Learning-based cybersecurity systems in practical environments. Issues such as data imbalance, model interpretability, and real-time deployment remain areas of ongoing research. This study builds upon existing work by applying Machine Learning techniques to a structured cybersecurity event dataset, focusing on improving detection accuracy and providing a practical approach for cyber attack classification.

V. METHODOLOGY

The proposed methodology follows a structured and systematic approach for detecting and classifying cyber attacks using Machine Learning techniques. It involves multiple stages, including data collection, preprocessing, feature engineering, model training, and performance evaluation.

Fig. 1 illustrates the overall workflow of the proposed system, providing a visual representation of the sequential steps involved in the implementation.

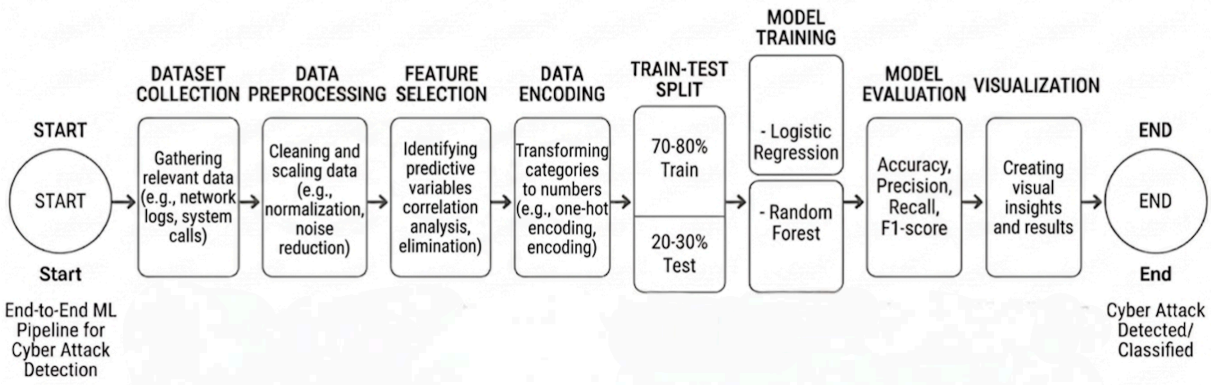

Fig. 1: Flowchart of Proposed Methodology

The flowchart presents a step-by-step representation of the methodology, starting from data collection and preprocessing, followed by model training and evaluation, and ending with performance analysis and visualization.

A. *Data Collection and Description*

The dataset used in this study is publicly available and can be accessed **here**. The dataset consists of structured cybersecurity event records containing attributes such as attack type, attack severity, data exfiltration status, threat intelligence, and response actions. These features provide information about system behavior and potential security incidents. The dataset includes multiple categories of cyber attacks, making it suitable for multi-class classification tasks.

The data was loaded using Python's pandas library, which allows efficient handling and manipulation of structured datasets. Initial inspection of the dataset was performed to understand its structure, data types, and distribution of values. This step is essential to identify inconsistencies and prepare the data for further processing.

B. *Data Preprocessing*

Data preprocessing was performed to ensure data quality and improve model performance. Missing values were handled by removing incomplete records using data cleaning techniques. This helps in maintaining consistency and avoiding errors during model training.

Irrelevant features such as timestamp, source IP address, destination IP address, and user agent were removed, as they do not provide meaningful information for predicting attack types. Additionally, the event ID attribute was excluded since it is a unique identifier and does not contribute to pattern recognition. Removing such features helps reduce noise and prevents misleading influence on the model.

C. *Data Encoding and Transformation*

Since the dataset contains categorical attributes, label encoding was applied to convert non-numeric data into numerical format. Machine Learning algorithms require numerical input; therefore, encoding is necessary for proper model training.

Each categorical column was transformed using encoding techniques provided by the scikit-learn library. This process ensures that the model can interpret categorical relationships effectively. The transformed dataset is then suitable for applying classification algorithms.

D. *Exploratory Data Analysis and Visualization*

Exploratory Data Analysis (EDA) was conducted to understand the distribution and characteristics of the dataset. Visualization techniques were used to gain insights into the data.

Bar graphs were plotted to analyze the distribution of attack types and attack severity levels. These visualizations help identify class balance and data patterns. Additionally, a correlation heatmap was generated to examine relationships between features. The analysis revealed weak correlations among features, indicating limited predictive relationships within the dataset.

E. *Feature Selection*

Feature selection was performed to retain only the most relevant attributes for model training. By removing redundant and less informative features, the dimensionality of the dataset is reduced, leading to improved computational efficiency.

This step also helps in enhancing model interpretability, as it focuses on features that contribute meaningfully to classification. The selected features were used as input variables, while the attack type was considered as the target variable.

F. *Data Splitting*

The dataset was divided into training and testing sets using an 80:20 ratio. The training dataset was used to train the Machine Learning models, while the testing dataset was used to evaluate their performance on unseen data.

This approach ensures that the models are assessed fairly and reduces the risk of overfitting. The train-test split was implemented using the train_test_split function from the scikit-learn library.

G. *Model Selection and Training*

Two Machine Learning classification algorithms were used in this study: Logistic Regression and Random Forest.

Logistic Regression was selected as a baseline model due to its simplicity and effectiveness in handling linear classification problems. It provides a reference point for evaluating model performance.

Random Forest was chosen as an ensemble-based algorithm capable of capturing complex and non-linear relationships in the data. It combines multiple decision trees to improve prediction accuracy and reduce overfitting.

Both models were trained using the training dataset. Logistic Regression was configured with an increased number of iterations to ensure convergence, while Random Forest was configured with multiple estimators and controlled depth to balance performance and complexity.

H. *Model Evaluation*

The performance of the trained models was evaluated using standard classification metrics, including accuracy, precision, recall, and F1-score. These metrics provide a comprehensive assessment of model performance by considering both correct predictions and classification errors.

A confusion matrix was generated to visualize the distribution of correct and incorrect predictions across different classes. Additionally, feature importance analysis was performed for the Random Forest model to identify the contribution of each feature in the prediction process.

The evaluation results help in understanding the effectiveness of the models and highlight the impact of dataset characteristics on classification performance.

VI. IMPLEMENTATION

A. *Development Environment*

The implementation of the proposed cyber attack detection system was carried out using the Python programming language due to its strong support for data analysis and Machine Learning applications. The development was performed in a Jupyter Notebook environment, which provides an interactive platform for executing code and visualizing outputs.

Several Python libraries were used during implementation. The pandas library was used for data handling and preprocessing, while NumPy was used for numerical operations. The scikit-learn library was used to implement Machine Learning models such as Logistic Regression and Random Forest, along with evaluation metrics [7]. Additionally, matplotlib and seaborn libraries were used for data visualization.

B. *Data Loading and Inspection*

The dataset was loaded into the system using the pandas library. Initial inspection was performed to understand the structure, data types, and completeness of the dataset.

Functions such as head() and info() were used to examine the dataset.

Fig. 2 presents a sample view of the dataset, illustrating the key attributes and their corresponding values used for cyber attack analysis. Fig. 3 presents the dataset structure, including the number of entries and data types of each column.

	Attack Type	Attack Severity	Data Exfiltrated	Response Action
0	Ransomware	Critical	False	Eradicated
1	Malware	Critical	True	Eradicated
2	Ransomware	High	False	Eradicated
3	DDoS	Critical	False	Contained
4	Malware	Medium	False	Eradicated

Fig. 2: Sample View of Processed Dataset

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 20000 entries, 0 to 19999
Data columns (total 10 columns):
 #   Column               Non-Null Count  Dtype
---  ------               --------------  -----
 0   Event ID             20000 non-null  object
 1   Timestamp            20000 non-null  object
 2   Source IP            20000 non-null  object
 3   Destination IP       20000 non-null  object
 4   User Agent           20000 non-null  object
 5   Attack Type          20000 non-null  object
 6   Attack Severity      20000 non-null  object
 7   Data Exfiltrated     20000 non-null  bool
 8   Threat Intelligence  20000 non-null  object
 9   Response Action      20000 non-null  object
dtypes: bool(1), object(9)
memory usage: 1.4+ MB
```

Fig. 3: Dataset Structure and Data Types

C. *Data Preprocessing Implementation*

Data preprocessing was performed to prepare the dataset for Machine Learning algorithms. Missing values were handled by removing incomplete records to maintain data consistency. Irrelevant features such as timestamp, source IP address, destination IP address, and user agent were removed, as they do not contribute to cyber attack classification.

Additionally, the event ID column was removed since it is a unique identifier and does not provide meaningful information for prediction. Removing such features helps reduce noise and improves model reliability.

Categorical variables present in the dataset were converted into numerical format using label encoding techniques. This step ensures compatibility with Machine Learning algorithms, which require numerical input data.

D. *Data Visualization*

Data visualization was performed to understand the characteristics of the dataset. Bar charts were used to analyze the distribution of attack types and attack severity levels. A correlation heatmap was also generated to examine relationships between features.

The visualization results indicated that the dataset has weak feature correlations, which affects the ability of the models to learn meaningful patterns. These visualizations are further discussed in the Results section.

E. *Model Implementation*

Two Machine Learning models were implemented in this study: Logistic Regression and Random Forest.

Logistic Regression was used as a baseline model due to its simplicity and efficiency in linear classification tasks. It provides a reference point for evaluating model performance.

Random Forest was implemented as the primary model due to its ability to capture non-linear relationships and handle complex data patterns. It combines multiple decision trees to improve prediction accuracy and reduce overfitting.

Both models were trained using the training dataset and were used to generate predictions on the testing dataset.

F. *Model Evaluation Implementation*

The performance of the models was evaluated using standard classification metrics, including accuracy, precision, recall, and F1-score. These metrics provide a comprehensive assessment of model performance.

A confusion matrix was generated to analyze prediction results across different classes. Additionally, an accuracy comparison graph and feature importance analysis were performed to evaluate and interpret model performance.

The detailed results and graphical analysis are presented in the subsequent Results and Discussion section.

G. *Implementation Workflow*

The overall implementation process follows a structured workflow:

- Load the dataset
- Perform data inspection and preprocessing
- Encode categorical variables
- Visualize data distribution and feature relationships
- Split the dataset into training and testing sets
- Train Machine Learning models
- Evaluate model performance using metrics and visualizations

This workflow ensures a systematic and reproducible approach for detecting cyber attacks using Machine Learning techniques.

H. *Tools and Technologies Used*

The implementation was carried out using Python due to its extensive support for data analysis and Machine Learning applications. Jupyter Notebook was used as the development environment for executing code and visualizing outputs. The pandas library was used for data manipulation, while NumPy was used for numerical computations. The scikit-learn library was used to implement Machine Learning models and evaluation metrics. Visualization libraries such as matplotlib and seaborn were used for generating graphical representations of the dataset and model performance.

VII. RESULTS AND DISCUSSION

A. *Model Performance Evaluation*

The performance of the implemented Machine Learning models was evaluated using standard classification metrics, including accuracy, precision, recall, and F1-score. Two models, Logistic Regression and Random Forest, were applied to classify cyber attack types using the processed cybersecurity dataset.

The results indicate that Logistic Regression achieved an accuracy of approximately 19%, while the Random Forest model achieved a slightly higher accuracy of around 20%.

Although Random Forest performed marginally better, both models exhibited relatively low classification performance. The classification reports show that precision and recall values vary across different classes, with some classes exhibiting lower values due to limited correct predictions.

B. *Data Distribution Analysis*

Fig. 4 presents the distribution of attack types in the dataset. The classes are nearly equally distributed, indicating a balanced dataset, which increases classification complexity when distinguishing features are weak.

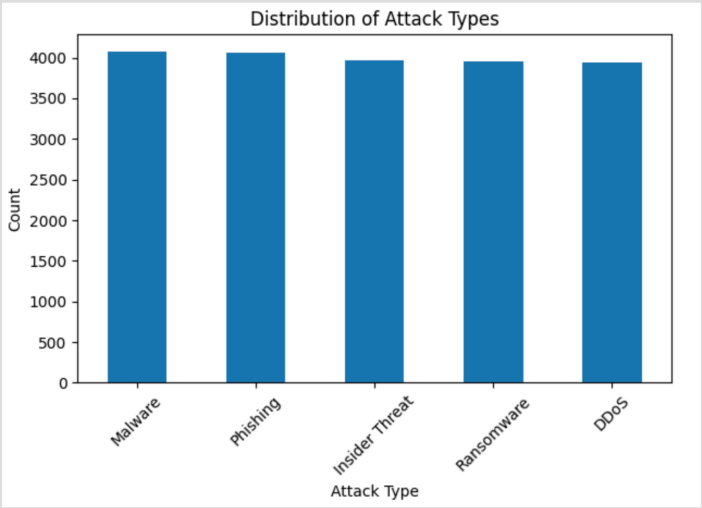

Fig. 4. Distribution of Cyber Attack Types

Fig. 5 illustrates the distribution of attack severity levels. The severity categories are also evenly distributed, further contributing to the difficulty in identifying clear patterns.

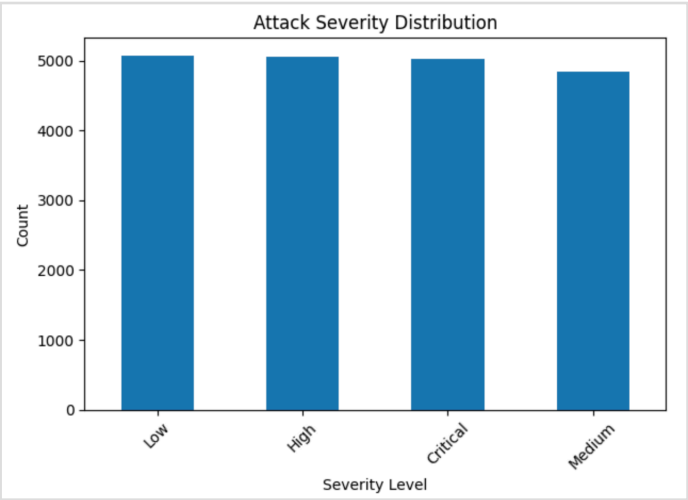

Fig. 5. Attack Severity Distribution

C. *Correlation Analysis*

Fig. 6 shows the feature correlation heatmap. The visualization indicates that most features have very weak correlations with each other, suggesting the absence of strong relationships within the dataset. This lack of correlation limits the ability of Machine Learning models to learn meaningful patterns, resulting in lower prediction accuracy.

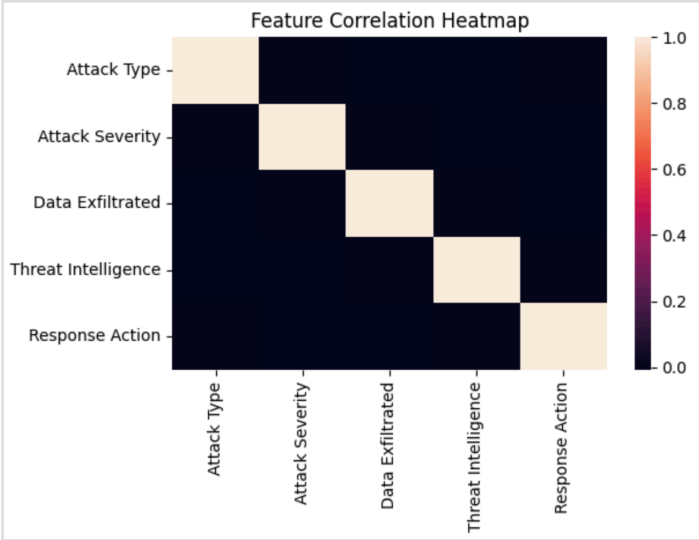

Fig. 6. Feature Correlation Heatmap

D. *Model Comparison*

Fig. 7 presents the comparison of model accuracies for Logistic Regression and Random Forest. It can be observed that the Random Forest model slightly outperforms Logistic Regression in terms of overall accuracy. This marginal improvement suggests that Random Forest, being an ensemble-based model, is better suited to capture non-linear relationships present in the dataset compared to the linear nature of Logistic Regression.

However, the improvement in performance is not significant, indicating that both models are similarly affected by the inherent limitations of the dataset. The weak feature correlations and lack of strong predictive attributes restrict the ability of both models to learn meaningful patterns. As a result, neither model is able to achieve high classification accuracy.

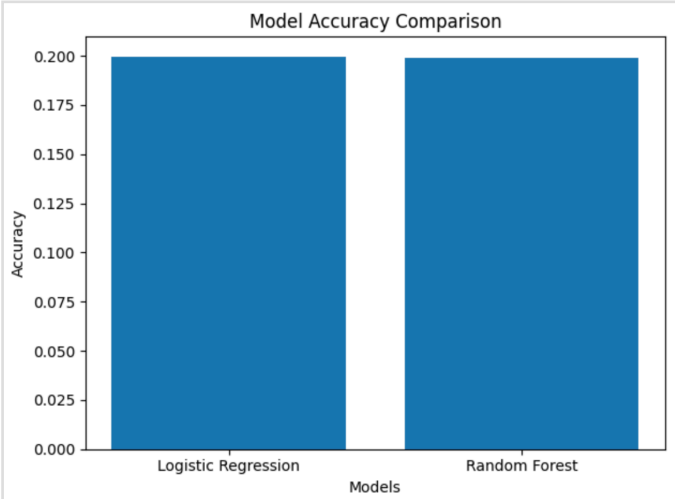

Fig. 7. Model Accuracy Comparison

E. *Confusion Matrix Analysis*

Fig. 8 illustrates the confusion matrix for the Random Forest model. The matrix shows that predictions are distributed across multiple classes, with no strong concentration along the diagonal. This indicates that the model frequently misclassifies instances, further confirming the challenge in distinguishing between different attack types.

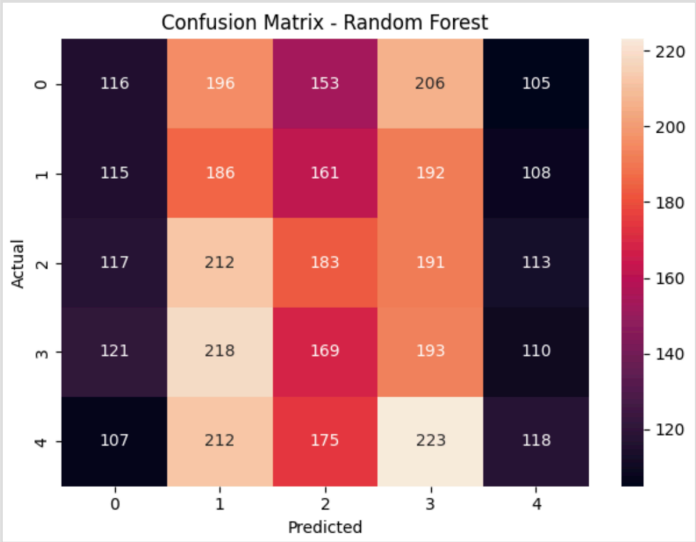

Fig. 8. Confusion Matrix of Random Forest Model

F. *Feature Importance Analysis*

Fig. 9 presents the feature importance analysis obtained from the Random Forest model. After removing irrelevant features such as event ID, the model highlights more meaningful attributes such as threat intelligence and attack severity. These features contribute relatively more to the model compared to other attributes, indicating that they carry some level of useful information for distinguishing between different cyber attack types. The improvement in interpretability after removing non-informative features demonstrates the importance of proper data preprocessing in Machine Learning workflows.

However, the overall distribution of feature importance values indicates that no single feature has a dominant influence on classification. Most features contribute only moderately or minimally, suggesting that the dataset lacks strong predictive attributes. This weak contribution from individual features limits the model's ability to form clear decision boundaries, resulting in lower classification accuracy. The analysis highlights that the effectiveness of Machine Learning models is highly dependent on the quality and relevance of input features, and emphasizes the need for improved feature engineering and more informative datasets in future work.

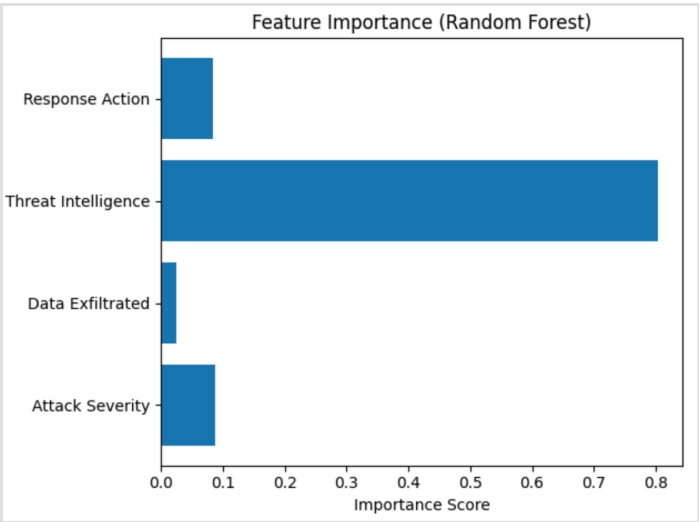

Fig. 9. Feature Importance Analysis

G. *Discussion*

The results demonstrate that both Machine Learning models face challenges in accurately classifying cyber attacks due to weak feature relationships and limited predictive information in the dataset. The correlation analysis confirms that the dataset lacks strong patterns required for effective classification.

Despite the low accuracy, the study successfully demonstrates the implementation of Machine Learning techniques for cybersecurity data analysis. The findings emphasize that the performance of Machine Learning models is highly dependent on the quality and relevance of input features rather than the choice of algorithm alone.

The analysis also highlights the importance of data preprocessing and feature selection. The removal of irrelevant attributes such as event ID improved model interpretability, although it did not significantly increase accuracy. Future improvements may involve advanced feature engineering, inclusion of more informative attributes, and application of more complex models to enhance classification performance.

H. *Class-wise Performance Analysis*

A detailed analysis of class-wise performance reveals that the models do not perform uniformly across all attack categories. Certain classes exhibit relatively higher recall, while others show very low or zero precision values. This indicates that the models are biased toward predicting certain classes more frequently while ignoring others.

The imbalance in prediction behavior can be attributed to the similarity of feature patterns across different attack types. Since the features do not provide strong distinguishing characteristics, the models struggle to form clear decision boundaries between classes. As a result, some classes are underrepresented in predictions, leading to undefined precision values.

I. *Impact of Feature Quality*

The performance of the models is significantly influenced by the quality and relevance of the input features. The correlation analysis demonstrated that most features have weak relationships with the target variable, indicating limited predictive capability.

Features such as threat intelligence and attack severity contribute relatively more to the model compared to other attributes. However, these features alone are not sufficient to achieve high classification accuracy. The absence of strong predictive features restricts the ability of the models to learn meaningful patterns, resulting in near-random prediction behavior.

J. *Comparison with Expected Performance*

In typical cybersecurity applications, Machine Learning models can achieve high accuracy when trained on datasets with strong feature relationships and well-defined patterns. However, in this study, the observed accuracy is relatively low due to the nature of the dataset.

The results suggest that the dataset may not contain sufficient discriminative information to support effective classification. This highlights the importance of dataset selection and feature engineering in Machine Learning-based cybersecurity systems.

K. *Limitations of the Study*

The study has certain limitations that affect model performance. One of the primary limitations is the lack of strong feature relationships within the dataset. Additionally,

the dataset may contain attributes that are either redundant or not directly relevant to the classification task.

Another limitation is the use of basic Machine Learning models without advanced feature engineering techniques. While the models used in this study provide a baseline understanding, more sophisticated approaches may yield improved results.

L. *Observations from Visual Analysis*

The graphical representations provide important insights into the dataset and model behavior. The distribution graphs indicate that the dataset is balanced across multiple classes, which increases classification complexity. The correlation heatmap confirms the absence of strong relationships between features, while the confusion matrix highlights frequent misclassifications across different categories.

The feature importance graph further supports these observations by showing that no single feature dominates the prediction process. These visual insights reinforce the conclusion that dataset characteristics play a critical role in determining model performance.

VIII. Conclusion and Future Scope

A. *Conclusion*

This study presented a data-driven approach for cyber attack detection using Machine Learning techniques applied to a structured cybersecurity event dataset. The implementation involved data preprocessing, feature selection, model training, and performance evaluation using classification algorithms such as Logistic Regression and Random Forest.

The results indicate that both models achieved relatively low accuracy, with Logistic Regression achieving approximately 19% accuracy and Random Forest achieving around 20%. Although Random Forest slightly outperformed Logistic Regression, the overall performance of both models remained limited. This behavior can be attributed to the characteristics of the dataset, which exhibits weak correlations among features and lacks strong predictive relationships with the target variable.

The analysis of graphical visualizations, including distribution plots, correlation heatmaps, and confusion matrices, further supports these findings. The balanced distribution of classes increases classification complexity, while the absence of strongly influential features restricts the ability of the models to learn meaningful patterns. Additionally, the removal of irrelevant features such as event ID improved model interpretability but did not significantly enhance prediction accuracy.

Despite the limitations in accuracy, the study successfully demonstrates the application of Machine Learning techniques in cybersecurity data analysis. The work highlights the importance of data preprocessing, feature selection, and proper evaluation in developing effective cyber attack detection systems. It also emphasizes that the performance of Machine Learning models is highly dependent on the quality and relevance of the dataset.

B. *Future Scope*

The results of this study suggest several directions for future improvement. One potential enhancement is the use of advanced feature engineering techniques to extract more meaningful information from the dataset. Incorporating additional features or transforming existing attributes may improve the ability of models to identify patterns in the data.

Another important area for future work is the use of more sophisticated Machine Learning and Deep Learning models, such as Gradient Boosting, Support Vector Machines, or Neural Networks. These models may be better suited for capturing complex relationships in cybersecurity data.

The dataset itself can also be improved by including more informative and relevant features that directly contribute to attack classification. Real-world datasets with richer attributes and stronger correlations may significantly enhance model performance.

Furthermore, future research can explore real-time cyber attack detection systems that continuously monitor network activity and respond dynamically to threats. Integrating Machine Learning models into live cybersecurity frameworks can improve automated threat detection and reduce response time.

Finally, addressing challenges such as class imbalance, noise in data, and feature redundancy can further improve the reliability and effectiveness of Machine Learning-based cybersecurity systems.

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